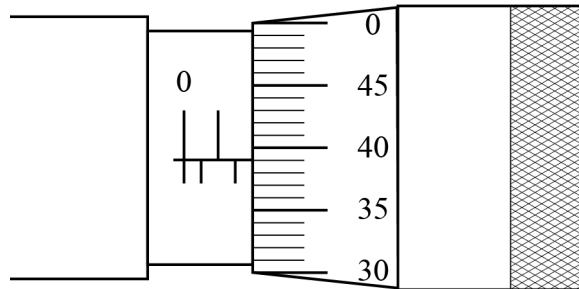


0 3

A student uses a micrometer, that has no zero error, to measure the diameter of a copper wire.

Figure 1 shows the micrometer scales as the diameter is measured.

Figure 1



0 3

1

Show that the cross-sectional area of the wire is about $2.8 \times 10^{-6} \text{ m}^2$

[2 marks]

0 3

2

The resistivity of copper is $1.7 \times 10^{-8} \Omega \text{ m}$
The wire has a length of 85 cm

Calculate the resistance of the wire.

Give your answer to an appropriate number of significant figures.

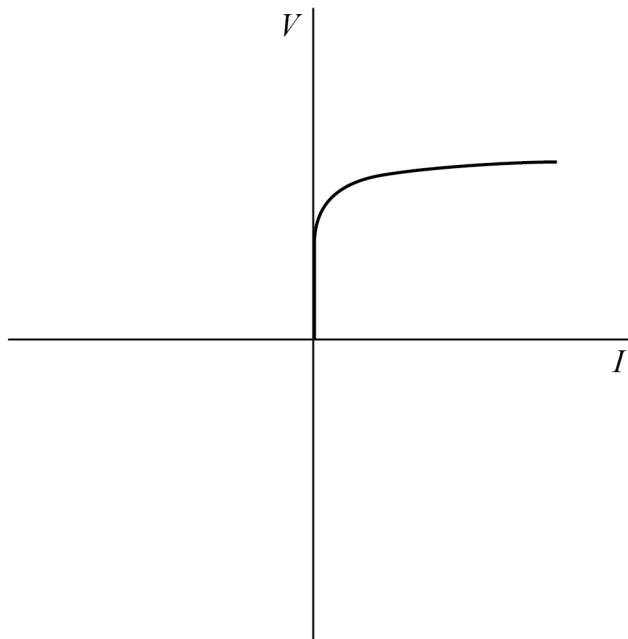
[3 marks]

resistance = _____ Ω

Don't write

Which component has the voltage–current (V – I) characteristics shown by the graph below?

[1 mark]



A filament lamp

☐

B metal conductor at constant temperature

☐

C semiconductor diode

☐

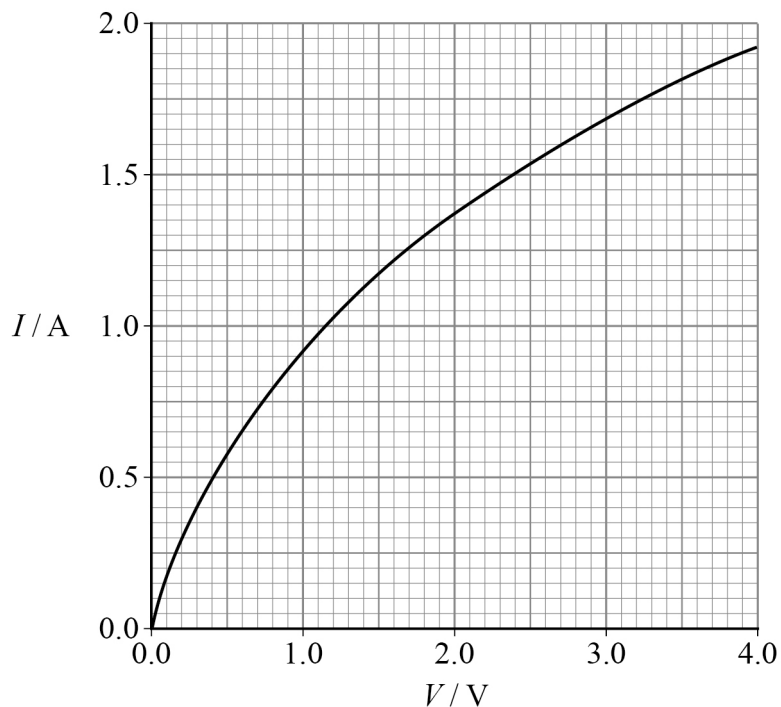
D thermistor

☐

0 5

Figure 2 shows the current-voltage (I - V) characteristic of a filament lamp.

Figure 2



0 5 . 1

Explain how **Figure 2** shows that the filament lamp is **not** an ohmic conductor.

[1 mark]

0 5 . 2

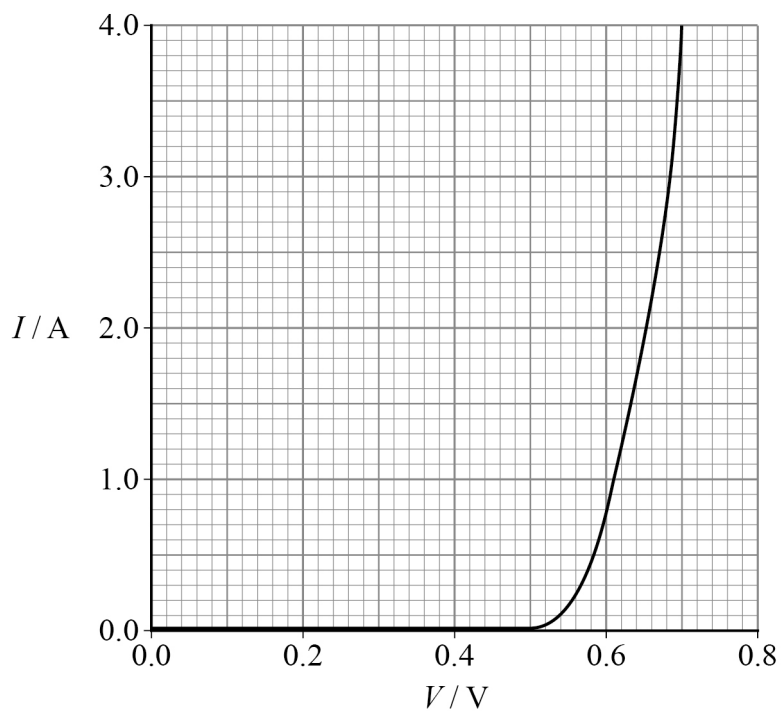
Explain, in terms of the particles in the filament, why the filament lamp is **not** an ohmic conductor.

[3 marks]



Figure 3 shows the I - V characteristic of a diode.

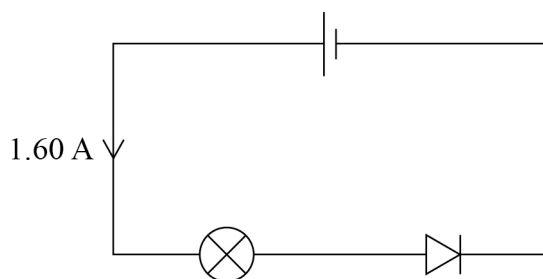
Figure 3



0 5 . 3

The filament lamp and the diode are connected in series with a cell of negligible internal resistance, as shown in **Figure 4**. The current in the circuit is 1.60 A.

Figure 4



Determine the emf of the cell.

[2 marks]

emf = _____ V

0	1
---	---

A metal has a resistivity of $4.9 \times 10^{-7} \Omega \text{ m}$.

0	1	.	1
---	---	---	---

Define resistivity.

[1 mark]

0	1	.	2
---	---	---	---

A wire resistor of resistance 6.8Ω is made from the metal.
The diameter of the wire is $2.4 \times 10^{-4} \text{ m}$.

Calculate the length of the wire.

[2 marks]

length = _____ m

0	1
---	---

Lightning strikes can happen when there is a large potential difference between a cloud and the ground.

The energy transferred during one lightning strike is 1.6×10^9 J.

A charge of 23.7 C moves between the cloud and the ground in a time t .

The magnitude of the current is 3.09×10^4 A.

Assume that the potential difference between the cloud and the ground is constant.

0	1	.	1
---	---	---	---

Calculate t .

[1 mark]

$t =$ _____ s

0	1	.	2
---	---	---	---

Calculate the potential difference between the cloud and the ground.

[1 mark]

potential difference = _____ V

0	1	.	3
---	---	---	---

A power station has an electrical output of 1300 MW.

Calculate the time taken for the power station to transfer 1.6×10^9 J.

[1 mark]

time taken = _____ s

3

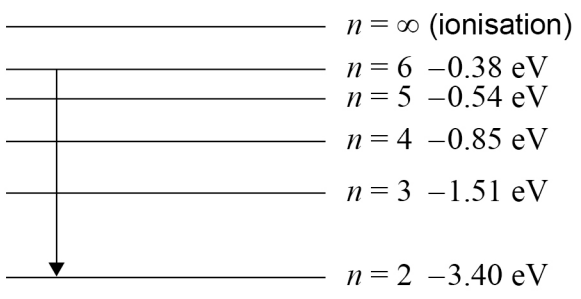


Do not write

0 5

Figure 5 shows the energy levels in a hydrogen atom.
The arrow indicates one transition.

Figure 5



not to scale

_____ $n = 1$ -13.6 eV ground state

0 5

1

State the energy associated with the $n = \infty$ (ionisation) level.

[1 mark]

0 5

2

The transition shown in **Figure 5** produces a photon of violet light.

Draw, on **Figure 5**, an arrow to represent a transition that would produce a photon in the ultraviolet part of the spectrum.

[1 mark]



0 5 . 3

Calculate the lowest frequency of a photon that can be emitted during any transition to the $n = 2$ level on **Figure 5**.

[4 marks]

frequency = _____ Hz

6

Turn over for the next question

Turn over ►



0 6 . 1 An electron is accelerated from rest through a potential difference V .

Show that the final momentum p of the electron is given by

$$p = \sqrt{2em_e V}$$

[3 marks]

0 6 . 2 The spacing between layers of atoms in a crystal is 2.0×10^{-10} m.

Estimate the momentum of electrons that will give an observable diffraction pattern of the crystal.

[2 marks]

momentum = _____ kg m s⁻¹



0 6 . 3 State the assumption that you made in Question **06.2**.

[1 mark]

0 6 . 4 Calculate the potential difference needed to accelerate electrons from rest so that they have the momentum that you estimated in Question **06.2**.

[1 mark]

potential difference = _____ V

7

Turn over for the next question

Turn over ►

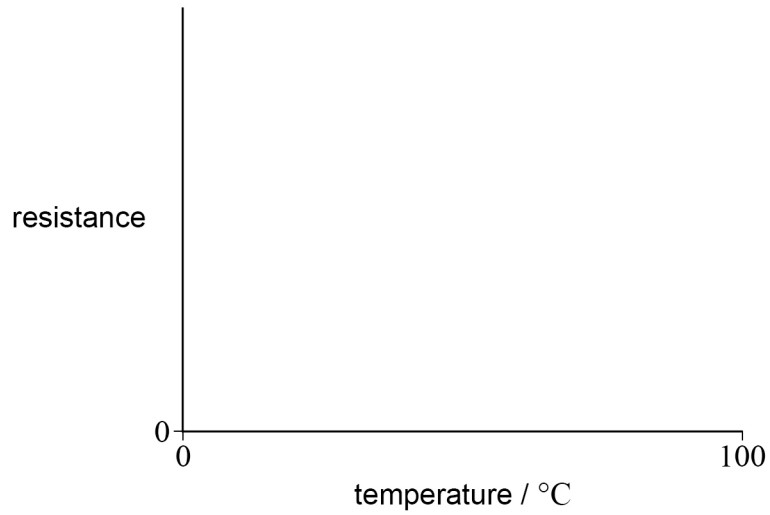


0 7 . 1

Sketch, on **Figure 6**, a graph to show the variation of resistance with temperature for a negative temperature coefficient thermistor for temperatures in the range of 0 °C to 100 °C.

[1 mark]

Figure 6



Your description should include:

- [5 marks]**

[illegible]

Do not write

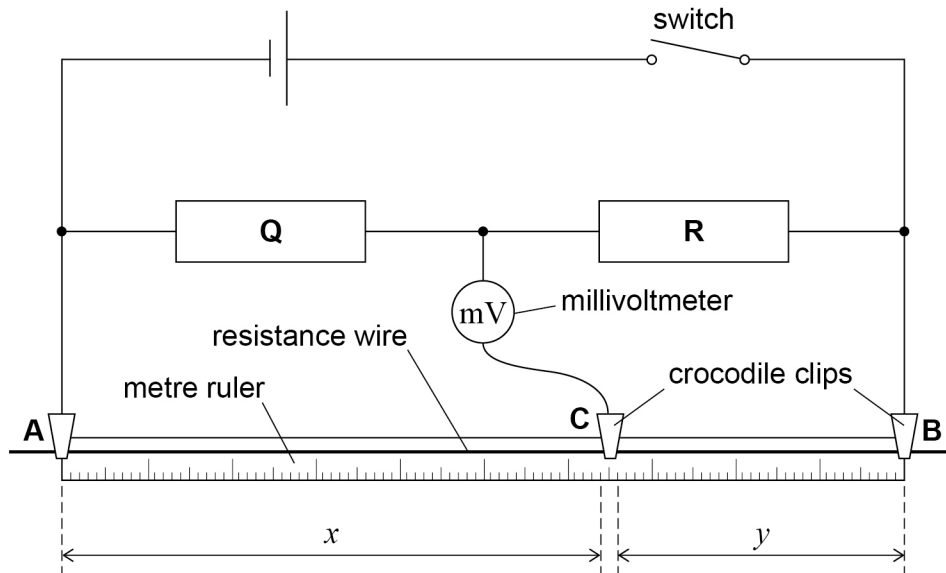
Figure 8 shows a circuit used to measure the resistance of a resistor **R**.

A metre of uniform resistance wire is attached to a metre ruler between crocodile clips **A** and **B**.

Crocodile clip **C** can be moved along the wire between **A** and **B**.

Q is a resistor of known resistance.

Figure 8



The switch is closed and crocodile clip **C** is moved along the resistance wire until the millivoltmeter reads zero.

09.1

Compare the potential difference (pd) across resistor **Q** with the pd across the wire between **A** and **C** when the millivoltmeter reads zero.

[1 mark]

Question 9 continues on the next page



A $4.7\ \Omega \pm 2\%$ resistor is chosen for **Q**.
The position of **C** is adjusted five times so that, each time, the millivoltmeter reads zero.
Each time, the length x of the wire between **A** and **C** and the length y of the wire between **C** and **B** are measured.

Table 1 shows the results.

Table 1

test	1	2	3	4	5
x / mm	539	541	537	540	538
y / mm	457	455	459	456	458
$\frac{x}{y}$	1.18	1.19	1.17	1.18	1.17

0 9 . 2 Show that the uncertainty in the mean value of $\frac{x}{y}$ is less than 1%.

[3 marks]



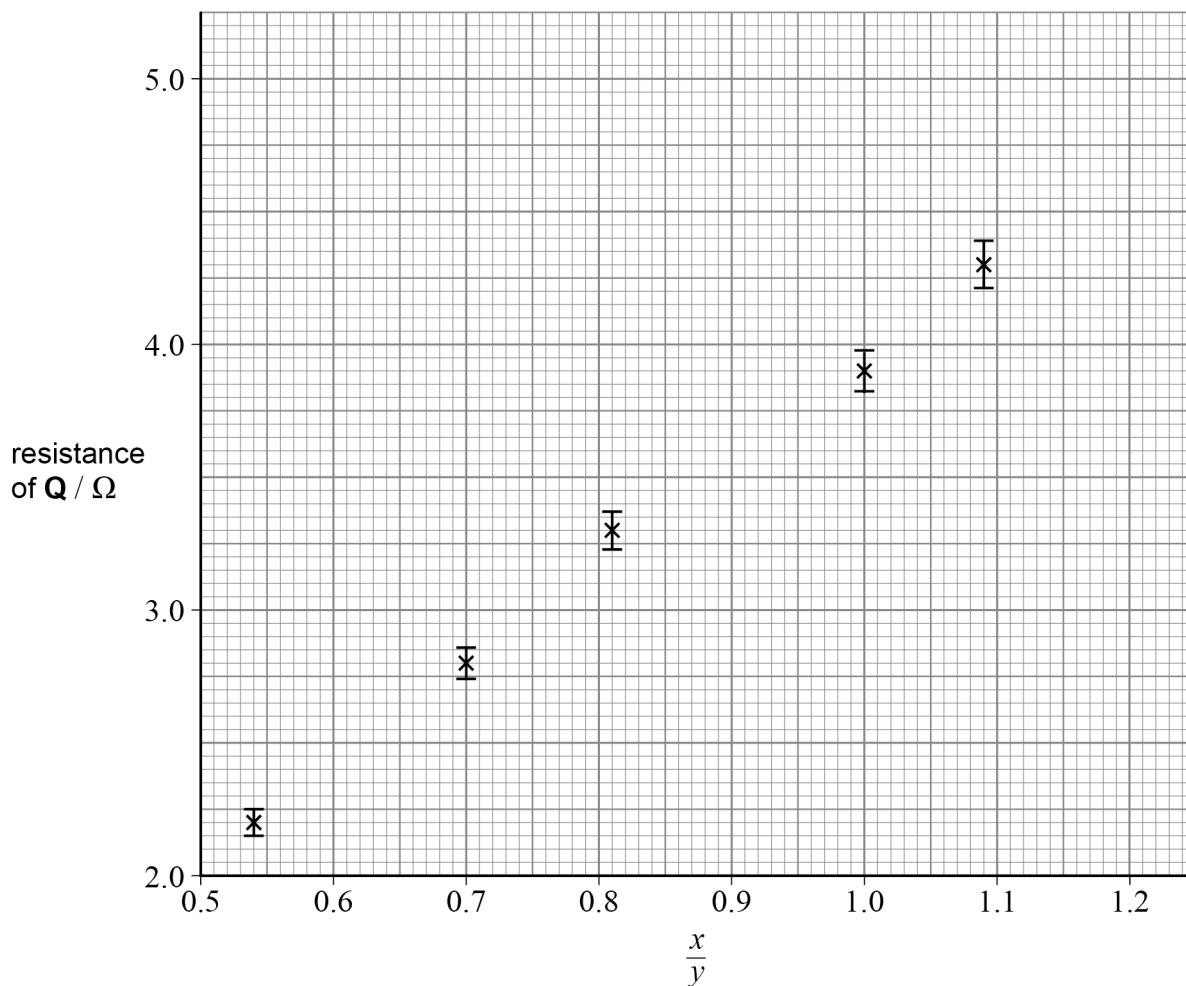
0 9 . 3

The experiment is repeated for five more values of resistor **Q**.
These data are plotted on **Figure 9**.

Plot, on **Figure 9**, the corresponding data point when the resistance of **Q** is $4.7\ \Omega$.
Draw an error bar to show the range of possible values of the resistance of **Q**.

[1 mark]

Figure 9



0 9 . 4

Draw a line of best fit on **Figure 9**.

[1 mark]

Question 9 continues on the next page

Turn over ►



0 9 . 5

The relationship between the resistances of **Q** and **R** is

$$\frac{\text{resistance of } \mathbf{Q}}{\text{resistance of } \mathbf{R}} = \frac{x}{y}$$

Determine, from your graph, the resistance of **R**.

[3 marks]

resistance of **R** = _____ Ω

9



Don't write

What is the fundamental (base) unit equivalent of the ohm?

[1 mark]

A $\text{kg m s}^{-2} \text{A}^{-2}$

☐

B $\text{kg m}^2 \text{s}^{-3} \text{A}^{-2}$

☐

C $\text{kg m s}^{-1} \text{C}^{-2}$

☐

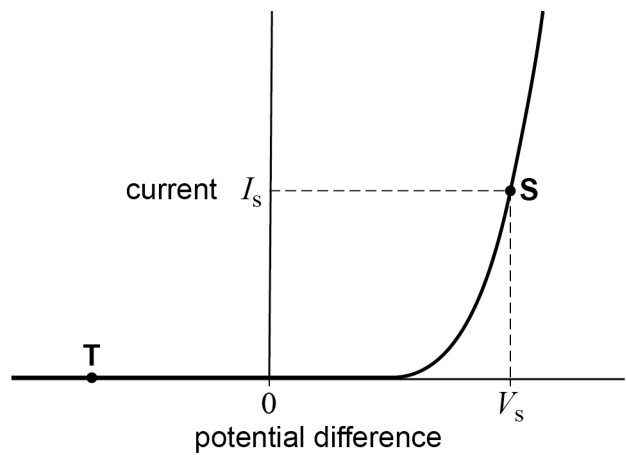
D $\text{kg m}^2 \text{s}^{-3} \text{C}^{-2}$

☐

Don't write

1 2

The graph shows the variation of current with potential difference for an ideal semiconductor diode.



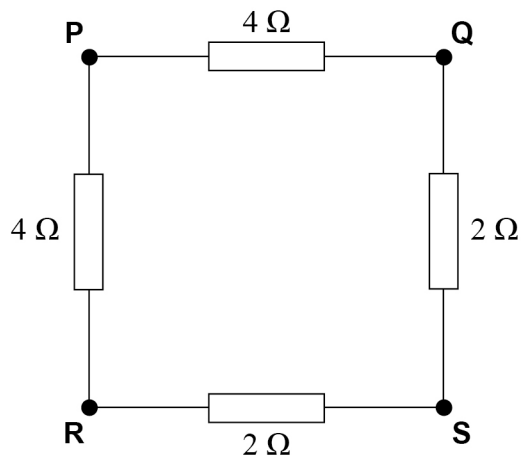
Which row gives the resistance of the semiconductor diode at the points marked **S** and **T**? **[1 mark]**

	Resistance at S	Resistance at T	
A	$\frac{V_s}{I_s}$	infinite	☐
B	$\frac{V_s}{I_s}$	zero	☐
C	the gradient of the graph at S	infinite	☐
D	the gradient of the graph at S	zero	☐

Turn over for the next question



Do not write



The resistance of the network shown is a minimum when connected across

[1 mark]

A P and S.

☐

B P and R.

☐

C Q and R.

☐

D Q and S.

☐

What are the resistances of ideal voltmeters and ideal ammeters?

[1 mark]

	Ideal voltmeter resistance	Ideal ammeter resistance	
A	infinite	zero	☐
B	infinite	infinite	☐
C	zero	zero	☐
D	zero	infinite	☐

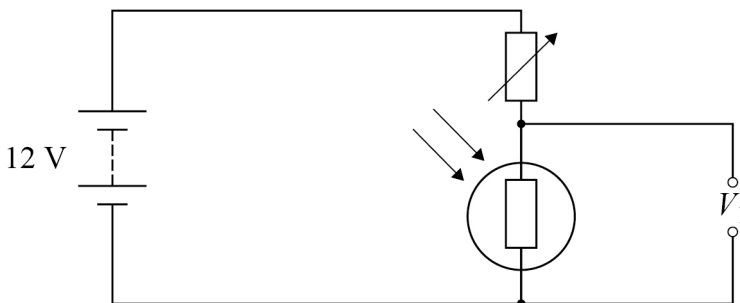


Don't write

Many cars have headlamps that switch on automatically when light levels become too low.

Figure 11 is a circuit diagram showing a light dependent resistor (LDR) being used as a light-detecting sensor. The output voltage V_1 from the circuit is used to control the headlamps.

Figure 11



The resistance of the variable resistor can be adjusted to change the light level at which the headlamps switch on.

The system is set to switch the headlamps on when the power of the light incident on the LDR is less than or equal to 12 mW .

The wavelength of the photons incident on the LDR is 550 nm .

Calculate the number of photons incident on the LDR in one second when the headlamps just switch on.

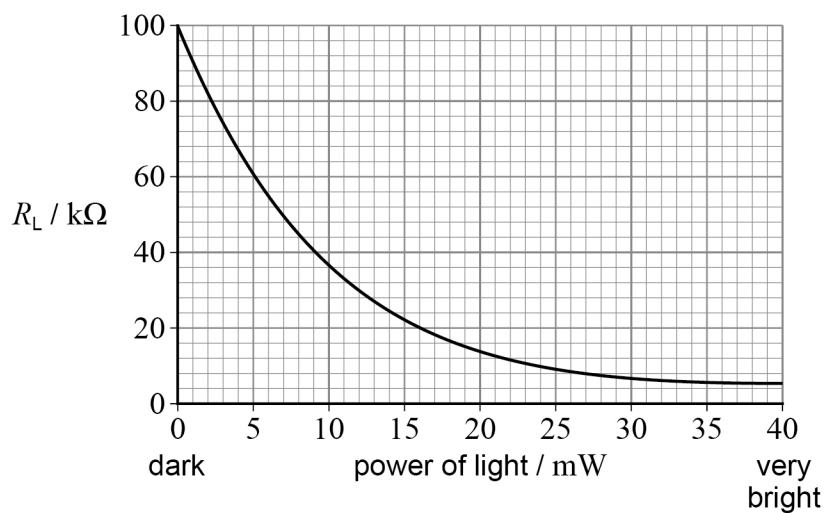
[3 marks]

number of photons in one second = _____

Question 10 continues on the next page

Figure 12 shows how the resistance R_L of the LDR varies with the power of the light incident on it.

Figure 12



1 0 . 2 V_1 must be at least 8.0 V for the headlamps to switch on.

Determine the resistance of the variable resistor for the headlamps to switch on when the power of the light is 12 mW.

[2 marks]

resistance = _____ Ω



1 0 . 3

An engineer wants to adjust the circuit so that the headlamps switch on when the light levels are greater.

State and explain what adjustment is necessary.

[3 marks]

8

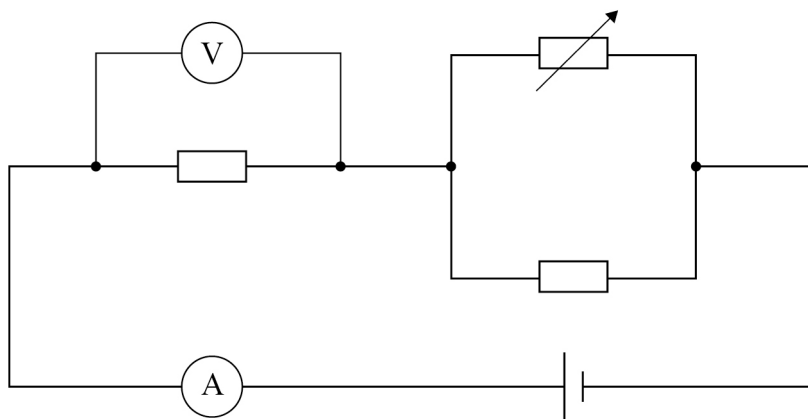
Turn over for the next question

Turn over ►



Do not write Do not write

A student notes the readings on the ammeter and voltmeter.



The resistance of the variable resistor is then decreased.

Which row shows the changes to the readings on the ammeter and the voltmeter?

[1 mark]

	Ammeter reading	Voltmeter reading	
A	decreases	decreases	☐
B	increases	increases	☐
C	increases	decreases	☐
D	decreases	increases	☐



Do not write Do not write

1

7

Do not write
outside the
box

A metal wire **X** has a resistance R .

A second wire **Y** of the same metal has twice the length and half the diameter of **X**.

What is the resistance of **Y**?

[1 mark]

A R

☐

B $2R$

☐

C $4R$

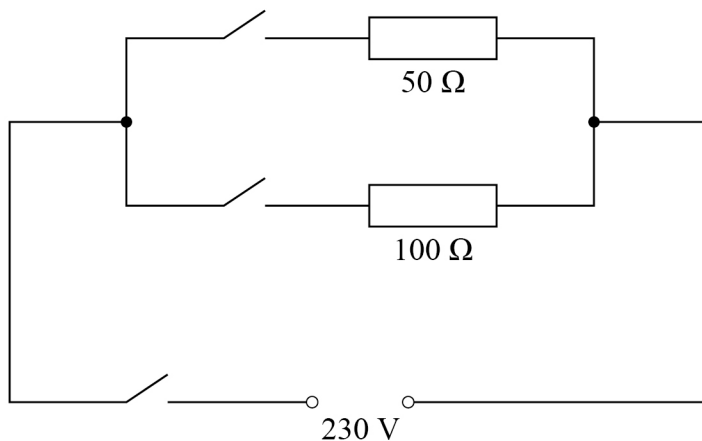
☐

D $8R$

☐

1 8

A heater consists of two resistors, one with a resistance of $50\ \Omega$ and the other with a resistance of $100\ \Omega$. The resistors are connected in parallel across a $230\ \text{V}$ supply. Each resistor can be switched on or off independently.



What is the maximum power available from the heater?

[1 mark]

A $530\ \text{W}$

☐

B $1100\ \text{W}$

☐

C $1600\ \text{W}$

☐

D $2100\ \text{W}$

☐

Don't write

2 **1** Which is **not** a unit for potential difference?

[1 mark]

A J C^{-1}

☐

B $\text{J s}^{-1} \text{A}^{-1}$

☐

C $\text{A } \Omega$

☐

D $\text{C } \Omega \text{ s}$

☐

Don't write

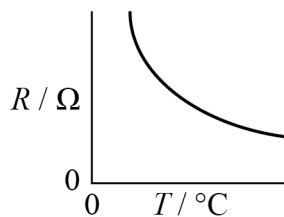
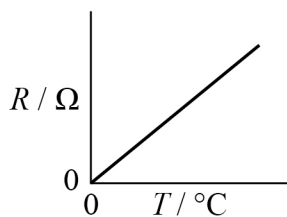
Which pair of graphs shows the variation of resistance R with temperature T for a metal wire and for a thermistor?

[1 mark]

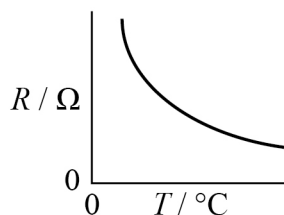
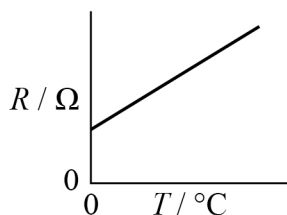
metal wire

thermistor

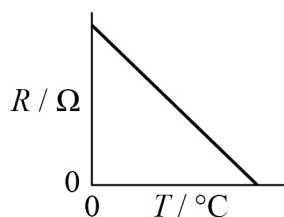
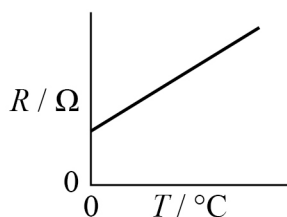
A


☐

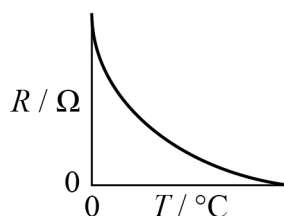
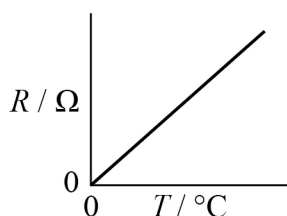
B


☐

C


☐

D


☐

Turn over for the next question

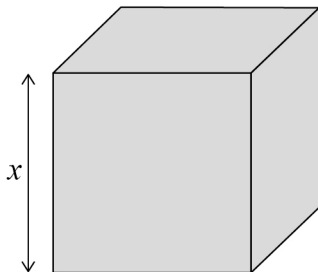
Turn over ►



Do not write Do not write

1 2

Each edge of a cube of conducting material has length x .



The resistance between any two opposite faces of the cube is R .

The same material is then reshaped into a uniform cylinder of length $4x$.

What is the resistance between the two opposite faces of this cylinder?

[1 mark]

A R

☐

B $4R$

☐

C $8R$

☐

D $16R$

☐

An electric current transfers energy at a constant rate of 54 J s^{-1} in an 18Ω resistor.

How many electrons pass a point in the resistor in 4.0 minutes?

[1 mark]

A 4.3×10^{19}

☐

B 6.5×10^{20}

☐

C 2.6×10^{21}

☐

D 4.5×10^{21}

☐

Do not write

2	0
---	---

A current of 2.0 A in a fixed resistor produces a power of 32 W .

What is the potential difference across the same resistor when a power of 64 W is being produced?

[1 mark]

A 16 V

☐

B 23 V

☐

C 32 V

☐

D 512 V

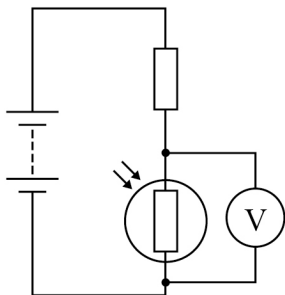
☐

Don't write

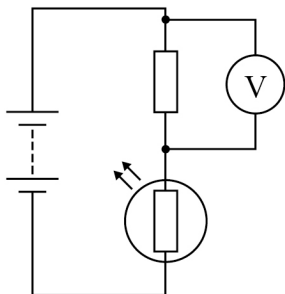
Which circuit shows an LDR arranged so that the voltmeter reading decreases as the light intensity increases?

[1 mark]

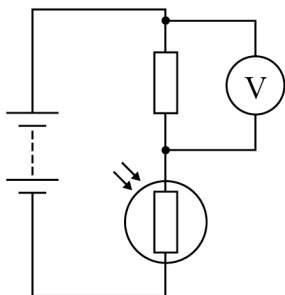
A



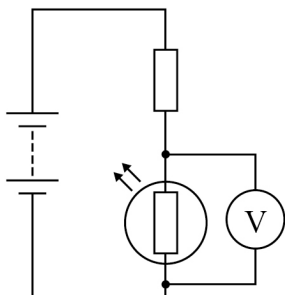
B



C



D



Turn over for the next question

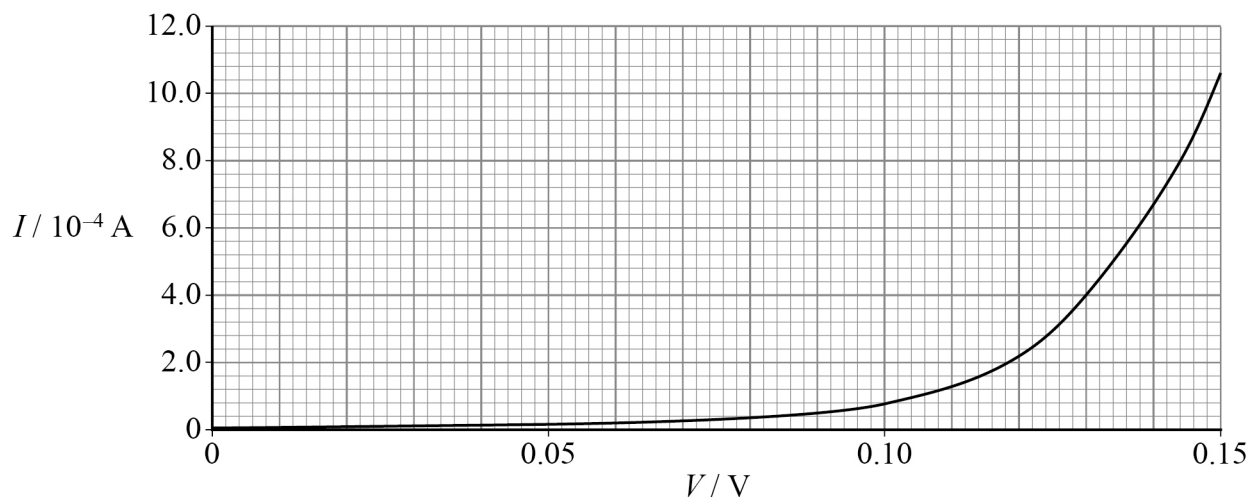
Turn over ►



Do not write

Figure 2 shows the I – V characteristic for a component.

Figure 2



Calculate the resistance of the component when the current in it is 0.80 mA.

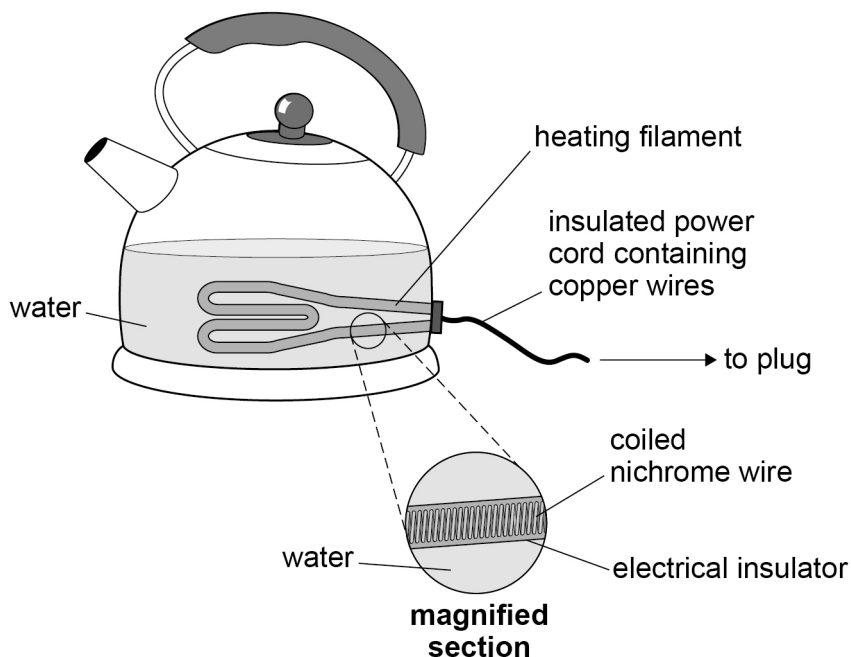
[2 marks]

resistance = _____ Ω

2

Don't write

Figure 13



This kettle contains a heating filament that consists of coiled nichrome wire surrounded by an electrical insulator. The heating filament is connected to the plug by a power cord that is made from copper wires surrounded by an electrical insulator.

When the kettle is switched on, energy is transferred from the filament to the water.

10.1

The resistance of the nichrome wire is $18\ \Omega$ at its operating temperature. The voltage across the nichrome wire is $230\ \text{V}$.

Calculate the power transferred by the nichrome wire.

[1 mark]

power = _____ W

Question 10 continues on the next page

The resistance of the copper wire in the power cord must be much less than the resistance of the nichrome wire.

1 0 . 2

Explain why.

[3 marks]

1 0 . 3

The resistivity of nichrome is $1.12 \times 10^{-6} \Omega \text{ m}$ at its operating temperature.
The resistance of the nichrome wire is 18Ω at this temperature.
The radius of the nichrome wire is 0.16 mm .

Calculate the length of the nichrome wire.

[2 marks]

length = _____ m



1 0 . 4

When the kettle is switched on, the temperature of the nichrome wire increases rapidly for about 5 ms, until the nichrome wire reaches its operating temperature. The temperature then remains constant.

Explain how the power consumption of the kettle varies for the first 10 ms. Calculations are not required.

[3 marks]

9

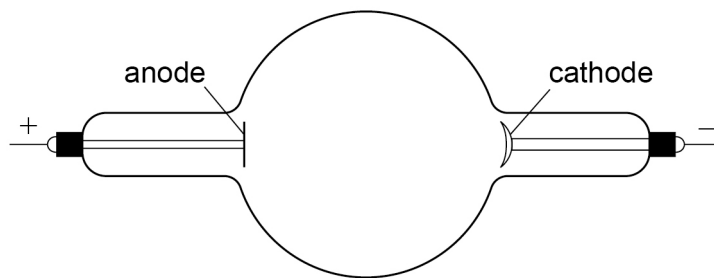
Turn over for the next question

Turn over ►



Don't write

A potential difference is applied across a neon gas-discharge tube.



In the tube, neon ions (Ne^+) with a charge of $+e$ travel from the anode to the cathode. Free electrons move in the opposite direction.

In 3.0 minutes, 4.4×10^{20} Ne^+ ions arrive at the cathode and 4.4×10^{20} free electrons reach the anode.

What is the total current between the cathode and the anode?

[1 mark]

A 0

☐

B 0.39 A

☐

C 0.78 A

☐

D 23 A

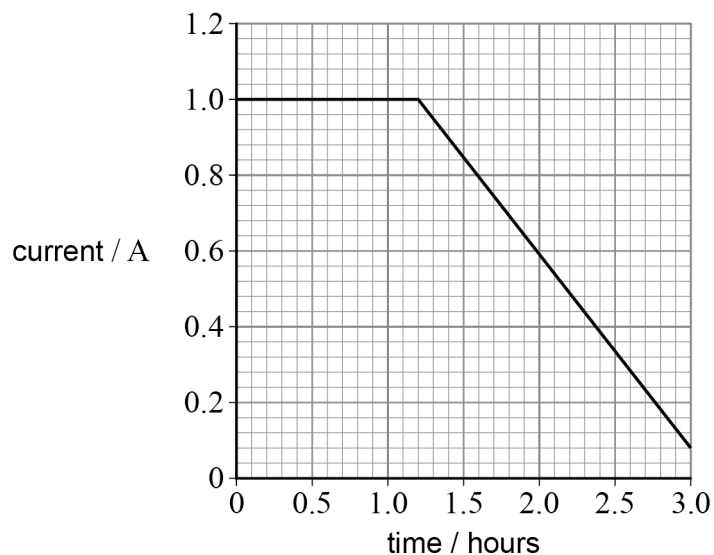
☐

Don't write

A cell provides an electric current for 3.0 hours.

Figure 1 shows the variation of the current with time.

Figure 1



Determine, in C, the charge transferred by the cell in this time period of 3.0 hours.

[3 marks]

charge transferred = _____ C

3

Turn over for the next question

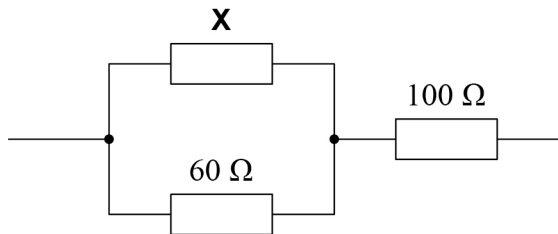
Turn over ►

Don't write

1 4

Do not write
outside the
box

The diagram shows a network of resistors.



The equivalent resistance of the network is $120\ \Omega$.

What is the resistance of **X**?

[1 mark]

A $60\ \Omega$ ☐

B $40\ \Omega$ ☐

C $30\ \Omega$ ☐

D $20\ \Omega$ ☐

1 2

When two resistors are connected together, the total resistance of the combination is $10\ \Omega$.

One resistor has a resistance of $12\ \Omega$.

What is the resistance of the other resistor?

[1 mark]

A $2.0\ \Omega$

☐

B $5.5\ \Omega$

☐

C $17\ \Omega$

☐

D $60\ \Omega$

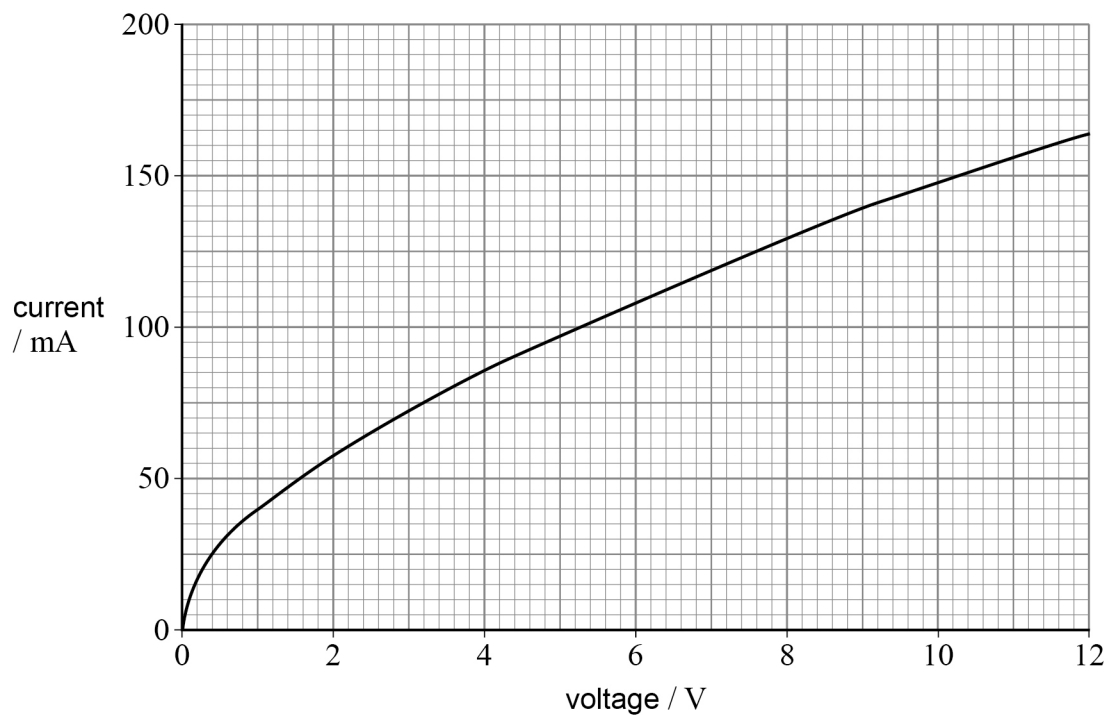
☐

Turn over ►



Do not write

The graph shows the I - V characteristic of a filament lamp **L**.



An identical lamp is connected in parallel with **L**.
The voltage across the parallel combination is 6.0 V.

What is the total power dissipated by the two lamps?

[1 mark]

A 0.42 W ☐

B 0.65 W ☐

C 0.90 W ☐

D 1.3 W ☐



Do not write

A superconducting component **S** is placed in series with a resistor of resistance R .

The voltage across the combination is V .

Which row is correct?

[1 mark]

	Current in S	Voltage across S	Voltage across the resistor
A	$\frac{V}{R}$	0	V
B	$\frac{V}{R}$	$\frac{V}{2}$	$\frac{V}{2}$
C	infinite	0	V
D	infinite	$\frac{V}{2}$	$\frac{V}{2}$

☐
☐
☐
☐

1 5

Two resistors R_1 and R_2 are connected in series.

The resistor values are:

$$R_1 = 14 \, \Omega \pm 20\%$$

$$R_2 = 30 \, \Omega \pm 10\%$$

What is the absolute uncertainty of the total resistance of the series combination?

[1 mark]

A $13 \, \Omega$

☐

B $9 \, \Omega$

☐

C $7 \, \Omega$

☐

D $6 \, \Omega$

☐

Turn over ►



Do not write

1 6

A potential difference is applied across a variable resistor.

The current in the variable resistor is I . The power dissipated by the variable resistor is P .

The resistance of the variable resistor is increased by 15%. The same potential difference is applied.

Which row shows the new current and the new power?

[1 mark]

	Current	Power	
A	$0.87I$	$0.87P$	☐
B	$0.87I$	$1.15P$	☐
C	$0.85I$	$0.85P$	☐
D	$0.85I$	$1.15P$	☐

0 2

A battery with a storage capacity of 1 A h can supply 1 A for one hour at its working potential difference.

A phone battery has a storage capacity of 4500 mA h.

0 2 . 1

Identify the SI unit for storage capacity.

Tick (✓) **one** box.

[1 mark]

A s⁻¹

☐

C

☐

J

☐

W

☐

0 2 . 2

Determine the storage capacity of the phone battery in the unit you chose in Question **02.1**.

[1 mark]

storage capacity = _____

2

Turn over for the next question

Turn over ►



Don't write

1**2**

Superconducting wires can transport large currents without any loss of energy. Power transmission systems do **not** yet use superconducting wires.

This is because it is too expensive to:

[1 mark]

A cool the superconducting wires.

☐

B generate the large currents.

☐

C manufacture the superconducting wires.

☐

D recycle the copper wires that are replaced.

☐

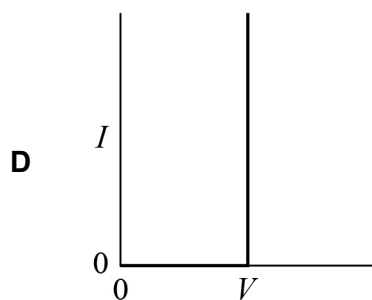
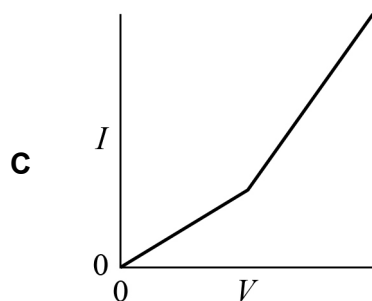
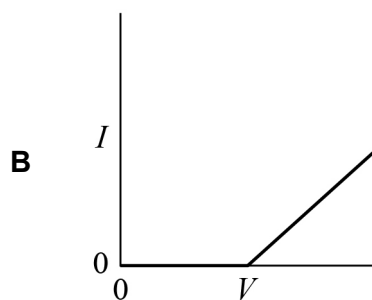
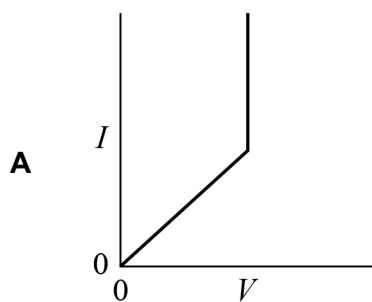
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1 3

An ideal semiconductor diode is connected in series with an ohmic resistor. A voltage is applied across the combination in the direction that allows current to flow.

What is the variation of current I with applied voltage V for the combination?

[1 mark]

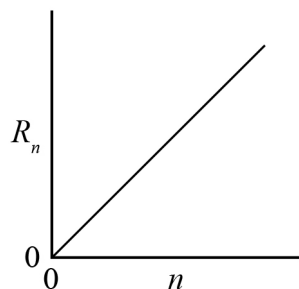
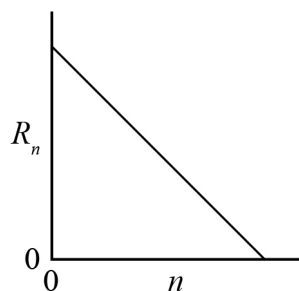
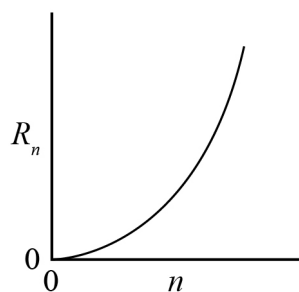
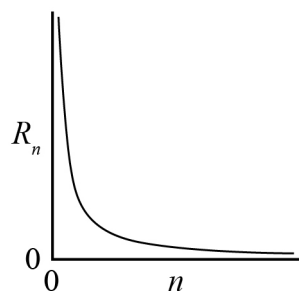


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1 4

A number n of identical resistors are connected in parallel. The overall resistance of the combination is R_n .

Which graph shows the variation of R_n with n ?

[1 mark]**A****B****C****D**

Do not write